Week 4 Presentation Content

Final dashboard, recommendations and reproducible submission

Presentation goal

Demonstrate the complete system, defend model quality and limits, and present evidence-backed actions that can be tested safely.

Deck map

Slide	Purpose
1	Final status and time review
2	End-to-end system demonstration
3	Final dashboard
4	Audience-segment findings
5	Model performance and governance
6	Content opportunities
7	Recommendations for promotion
8	Recommendations for testing and controls
9	Final submission and reproducibility
10	Limitations, reflection and Q and A

Evidence to keep open during Q and A

- dashboard/ for the Power BI project and semantic model
- MySQL writeback tables and dashboard views in sql/
- report/ and output/ for the final paper and deliverables

How to use this file Copy the on-slide text into PowerPoint. Use the visual direction to build the graphic, and rehearse from the speaker cue. Update blank actual-effort fields before presenting.

Slide 1 Week 4 closes the analytical and reporting loop

On-slide copy

- WHAT: demonstrate the full pipeline, final dashboard, recommendations, paper and reproducible repository.
- WHEN: freeze the model version and dashboard data before the final presentation.
- WHY: the final assessment requires one defensible chain from source data to decision.
- PLANNED EFFORT: 16-20 team hours. ACTUAL EFFORT: _____ hours. GAP: _____ hours.
- ACTION FROM GAP: _____

Visual direction Use a completion checklist with one evidence link per deliverable.

Speaker cue State the final model version and data refresh time at the start of the live demo.

Slide 2 The system runs from raw files to governed decisions

On-slide copy

- MovieLens files are validated and loaded into MySQL.
- KNIME reads curated tables, creates leakage-safe features and runs K Means plus Logistic Regression.
- Segments, predictions and metrics are written back to versioned MySQL tables.
- Power BI reads dashboard views and exposes evidence through filters and five pages.
- Recommendations use model evidence only after coverage, support and limitations are checked.

Visual direction Use tmp/report/figures/architecture.png. Animate one stage at a time if presenting in PowerPoint.

Speaker cue Demonstrate one traceable record or KPI across the layers. The purpose is proof of integration, not a tour of every script.

Slide 3 Five dashboard pages separate decisions from governance

On-slide copy

- Executive Overview answers whether the project has usable evidence and where action is concentrated.
- Audience Segments explains who the model groups and how those groups differ.
- Positive Engagement Model shows accuracy, discrimination, errors and coverage.
- Content Opportunity Matrix compares predicted appeal, actual response, historical popularity and support.
- Model Governance records versions, thresholds, split logic and limitations.
- Filters for model version, segment, genre and minimum support keep comparisons consistent.

Visual direction Use tmp/report/figures/dashboard_map.png, then switch to the live PBIP dashboard.

Speaker cue Show the same filter path across pages so the audience sees how one question becomes a governed decision.

Slide 4 Segments support different curation strategies

On-slide copy

- Genre Specialists are the largest segment and have the highest positive-rating rate at 71.43%.
- Selective Critics provide the strongest title-level support because they average 385.3 TRAIN ratings per user.
- Diverse Explorers show relative thriller and action affinity.
- Mainstream Enthusiasts show relative family and animation affinity but the lowest positive rate at 42.06%.
- K=4 was retained for interpretability even though K=5 had a slightly higher silhouette.

Visual direction Use the segment profile panel from tmp/report/figures/clustering_summary.png.

Speaker cue Translate profiles into curation hypotheses. Do not treat the labels as fixed identities or causal explanations.

Slide 5 The engagement model adds lift but has narrow TEST coverage

On-slide copy

- Accuracy is 69.26%, 9.93 percentage points above the 59.32% baseline.
- AUC is 75.83%; F1 is 64.92%; precision is 60.58%; recall is 69.94%.
- The model evaluates 76,633 TEST rows, equal to 5.11% coverage.
- Cold-start exclusions account for 94.89% of TEST rows, so the model is a warm-start decision aid.
- Governance controls: chronological splits, TRAIN-only aggregates, minimum support and model-version filters.

Visual direction Use tmp/report/figures/model_evaluation.png with a large 5.11% coverage label.

Speaker cue Answer performance questions with both quality and applicability: how well the model ranks covered rows, and how many rows it covers.

Slide 6 The opportunity matrix distinguishes promotion from overexposure

On-slide copy

- Only Selective Critics has title-segment pairs with support >= 50: 107 qualifying pairs.
- Hidden opportunities include Hotel Rwanda, My Neighbor Totoro, City of God and Oldboy.
- The Incredibles has high predicted appeal but a lower actual positive rate, so it needs testing rather than automatic promotion.
- Overexposed candidates include Home Alone, Star Wars Episode I, Twister and The Mask.
- Support thresholds prevent unstable small samples from driving the recommendation list.

Title	Support	Predicted	Actual	TRAIN popularity
Hotel Rwanda	108	76.72%	77.78%	215
My Neighbor Totoro	58	74.87%	74.14%	487
City of God	60	71.66%	70.00%	968
Oldboy	52	69.94%	67.31%	72

Visual direction Use tmp/report/figures/content_opportunity.png and label the four action quadrants.

Speaker cue Explain that opportunity is a prioritization signal. It does not prove that a campaign will cause higher engagement.

Slide 7 Promote validated hidden opportunities to Selective Critics

On-slide copy

- Recommendation 1: test Hotel Rwanda and My Neighbor Totoro in Selective Critics placements because predicted and actual positive rates both exceed 74% with support above 50.
- Recommendation 2: add City of God and Oldboy to discovery rows where low historical popularity is the main constraint.
- Primary measure: positive-rating rate among eligible warm-start users; guardrails: minimum support, exposure count and title-level complaint or hide rate if available.
- Owner: content strategy with analytics review. Decision window: one controlled campaign cycle.

Visual direction Use two recommendation cards only in the slide deck, each with evidence, owner and test measure.

Speaker cue State why these titles qualify: high predicted appeal, aligned actual response, sufficient support and comparatively low popularity.

Slide 8 Use experiments for uncertain titles and reduce overexposure

On-slide copy

- Recommendation 3: A/B test The Incredibles because predicted appeal is 74.45% but actual positive response is 60.75%.
- Recommendation 4: reduce default placement of Home Alone, Star Wars Episode I, Twister and The Mask for this segment; test alternative discovery titles instead.
- Randomize eligible users, keep placement and timing consistent, and compare positive response with confidence intervals.
- Do not generalize beyond warm-start users or claim revenue impact without exposure and commercial outcome data.

Visual direction Use a simple test design: eligible users -> random assignment -> content variants -> measured response.

Speaker cue Frame reduction as reallocation within one segment, not a global removal decision.

Evidence Home Alone actual positive rate 17.65%; Star Wars Episode I 19.61%; Twister 10.17%; The Mask 5.66% for qualifying Selective Critics pairs.

Slide 9 The final submission is reproducible from repository assets

On-slide copy

- Repository contains setup guidance, SQL scripts, KNIME documentation, Power BI project files, evidence outputs and the report.
- The IEEE-style paper covers Abstract, Keywords, Introduction, Related Work, Methodology, Experiments and Results, Discussion and Recommendations, Conclusion and References.
- The paper should remain within 6-10 pages and cite at least five credible sources.
- Before submission, confirm clean credentials, relative paths, model version consistency and a fresh dashboard refresh.
- Archive the exact inputs and outputs used in the live demo.

Visual direction Use a repository tree with check marks beside reproducibility-critical folders.

Speaker cue Name the command or workflow sequence needed to rebuild the outputs. Avoid saying only that the repository is complete.

Slide 10 The conclusion keeps decisions within the evidence

On-slide copy

- What worked: end-to-end integration, out-of-time evaluation, interpretable models and decision-oriented writeback.
- What remains limited: historical data, no exposure or revenue outcomes, overlapping clusters and low warm-start TEST coverage.
- Next step: run controlled experiments on qualifying opportunities and monitor coverage before expanding use.
- PLAN VERSUS ACTUAL: planned _____ hours; actual _____ hours; gap _____ hours.
- Team reflection and corrective action: _____
- Q and A backup: data quality, split dates, leakage controls, K choice, confusion matrix, coverage and recommendation thresholds.

Visual direction End on one summary diagram: evidence -> bounded decision -> experiment -> learning.

Speaker cue Invite questions on the evidence chain. If asked about business impact, distinguish model performance from causal campaign lift.